Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/10/2005
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Project Title:	Mechanisms and Functional Consequences of Protein Kinase C Isoform Translocation Inhibition in Monocytes Exposed to Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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Comments:	PI is deceased as of February 2021		
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hatton, Jason (NCIRE)		
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	This study is aims to investigate the mechanisms by which microgravity affects protein kinase C (PKC) signaling in monocytes and the functional consequences of these changes on downstream signaling events, gene expression and cell fate. When protein kinase C isoforms are activated they translocate to the cell membrane, which is an key step in signal transduction by these kinases. Previous spaceflight experiments have shown that protein kinase C isoform translocation in monocytes & lymphocytes is inhibited in microgravity, with individual isoforms being differentially affected. The objectives of this research project are: (1) Fully characterize the effect of microgravity on, (a) the activation of key PKC isoforms (b) spatial distribution / redistribution of PKC isoforms within the cell and (c) signaling downstream of PKC in response to mitogenic stimulation. (2) Determine the effect of microgravity on PKC regulated genes which control monocyte differentiation, the initiation of apoptosis and cell cycle arrest (3) Determine if it is possible to restore agonist induced activation of individual PKC isoforms inhibited by microgravity by application of PKC isoform specific		

Task Description:	translocation enhancers and in turn determine whether these agents can restore expression of genes perturbed by microgravity. The experimental approach consists of stimulating PKC isoform activation in U937 cells using differentiation inducing agonists in microgravity and a 1.g on-board centrifuge. Gene chips will be used to identify the changes in the pattern of gene expression between microgravity, to identify which PKC isoform signaling pathways are affected by microgravity. Reverse transcriptase – polymerase chain reaction (Rt-PCR) will be used to confirm expression of key genes identified from the gene array data. The sub-cellular localization of PKC isoforms and expression of key surface activation markers will be determined in formaldehyde fixed samples. Collectively, these data will allow us to explore in detail the consequences of perturbed PKC signaling in microgravity at the level of gene expression and cell fate. We plan to analyze the expression of patterns of gene expression induced by differentiation inducing agonists in the U937 human monocyte cell line in microgravity and 1.g, to determine the functional effects of alterations in PKC signaling that occur in microgravity. The objectives of this research project are: (1) Fully characterize the effect of microgravity on, (a) the activation of key PKC isoforms (b) spatial distribution / redistribution of PKC isoforms within the cell and (c) signaling downstream of PKC in response to mitogenic stimulation. (2) Determine the effect of microgravity on PKC regulated genes which control monocyte differentiation, the initiation of apoptosis and cell cycle arrest (3) Determine if it is possible to restore agonist induced activation of individual PKC isoforms inhibited by microgravity by application of PKC isoform specific translocation enhancers and in turn determine whether these agents can restore expression of genes perturbed by microgravity. Collectively, these data will allow us to explore in detail the mechanisms and consequences of perturbed PKC signaling in microgravity at the level of gene expression and cell fate.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study investigates the mechanisms by which microgravity affects protein kinase C (PKC) signaling in monocytes and the functional consequences of these changes on downstream signaling events, gene expression and cell fate. This is important since many earthbound biological processes depend on this pathway. We plan to analyze the expression of patterns of gene expression induced by differentiation inducing agonists in the U937 human monocyte cell line in microgravity and 1.g, to determine the functional effects of alterations in PKC signaling that occur in microgravity as well as on earth.
Task Progress:	Expression of key monocyte differentiation genes: We have made & tested PCR primers for key genes in monocyte differentiation. These include the cytokines TNF-alpha, IL-1beta, IL-1alpha & cell cycle regulators p21WAF and cyclin E. Induction of expression of these genes, as well as the early immediate genes c-fos, egr-1 and COX-2 are shown below. As expected early immediate genes are expressed transiently at 30 minutes after stimulus, whereas cytokine genes are expressed 3-6h after stimulus. Expression of p21WAF, an important regulator of the cell cycle in monocytes, begins at 6h after stimulus. These data are important for selecting the timepoints for sampling mRNA expression in the flight experiment Flow Cytometry: Flow cytometry was used to assess cell activation and cell cycle progression in a population of samples. We have tested flow cytometry protocols to quantify expression of early activation markers, including CD11c (an adhesion molecule, important in monocyte function during differentiation) and CD13 (a monocyte differentiation marker). Additionally we have tested propidium iodine staining as a method for quantifying the number of cells in each phase of the cell cycle using flow cytometry. Initial definition of the experiment for flight in the ESA Biolab facility was performed in July 2003. A preliminary experiment definition document and experiment flow was generated by Dr Enno Brinckmann at ESA/ESTEC based on our experiment proposal. We then refined this based on our preliminary data & current experiment design. The current experiment flow consists of 12 experiment cassettes, with 6 cassettes on the 1.g centrifuge & 6 cassette in microgravity, as described in figure 4 below. This experiment flow best accommodates the objectives of the experiment within the capabilities and constraints of the Biolab facility and was developed based on data from our preliminary experiments. Additionally the flight experiment was named the “PKINASE”. There are several important tasks that need to be completed in the next year to prepare the experiment for flight in the ISS Biolab facility. The Biolab will be in the ESA Columbus module on station. To ensure success it is important that all preparations and test be performed well ahead of the flight, since this is a new facility and ISS is a very different environment to the Space Shuttle which was used for all our previous experiments. Specifically, (1) Refine experiment definition & protocol, based on inputs from further test experiments. These data are critical for defining our flight hardware requirements. Specifically; (a) Develop protocol for unfreezing & culture of U937 cells in flight experiment. A protocol to do this reliably within the constraints of Biolab / ISS needs to be developed. We have a routine laboratory protocol for U937 cell unfreezing & culture and have performed some preliminary tests of possible freezing / unfreezing protocols with MC3T3-E1 bone cells. However, we need to test a variety of methodologies for preparation of the cells for freezing & then subsequent unfreezing in-flight. Data from these tests is crucial for the design of the cell culture / experiment cassette, overall experiment design and timeline. (b) Test stability of fixed & frozen samples, sample fixation methodologies as well as storage of reagents. The longer duration of an ISS missions (3-6 months), compared to a Shuttle flight, requires validation of the sample fixation methodologies, with storage times simulating the time until sample recovery post-flight. Furthermore, we need to optimize the fixation protocols and define the minimum sample volume required for each analysis methodology - again this is a driver for experiment cassette design, as well as the overall experiment flow. (c) Continued ground based experiments to refine endpoints for use in flight experiment (e.g. surface activation markers, microscopy) & provide baseline data for interpretation of flight experiment results. These data are drivers for selection / development of our flight experiment cassettes. Specific tests which need to be performed include; - Selection of the best surface markers for monocyte activation for use in cytofluorimetry analysis & determination of the timecourse of expression to determine optimal timepoint for sampling. - Fluorescence microscopy of PKC translocation and activation, with optimization of fixation & staining protocols. Kinetic experiments of PKC activation following agonist stimulation to determine optimal “early” and “late” timepoint for flight experiment. (d) Test experimental protocol in flight hardware, optimize fixation protocols & sample volume / quantity budgets for analysis. This is important to identify and remedy any potential problems with the hardware ahead of the flight. We need to optimize the science gain from the limited sample volume and number we will have from the flight experiment. Based on data from (b & c) we will plan out how the sample is divided between the different analysis methods (sample budget). Furthermore, once experiment hardware is available we need will need to test how the cells will grow within the hardware & test the experiment protocol in the hardware. Based on our previous experience (MHF, 4 Shuttle experiments; JPH 4 Shuttle experiments), testing &

	troubleshooting problems with hardware is time consuming. Therefore, it is important that we start this well ahead of the flight. (2) Optimize Gene array analysis for use in the flight experiment and baseline ground experiments. Gene chip analysis will allow us not only to achieve the original objectives of the experiment, but also to expand the scope of the study considerably. In the two years since the experiment was proposed we have gained experience in gene chip analysis, primarily in preparation of the STS-107 Leukin experiment. The specific advantages of using gene chip analysis are the following: · Gene chips allow us to examine changes in the expression of a large number of genes over time, rather than just a small selection. An example of a recent experiment by our laboratory using gene chips to measure T-cell activation in an altered gravity environment is given in Figure 5. · Ground based experiments with gene chips using inhibitors against specific signaling pathways will allow us to determine which pathways regulate which genes. This will provide us with reference data to compare with the flight data to determine which signaling pathways are affected by microgravity. · By assessing gene expression at multiple timepoints we obtain data for clustering analysis which can help identify the signaling pathways which are affected by microgravity. This also allows us to pinpoint the timescale that the cells are sensitive to microgravity. We plan to run a full baseline experiment of flight experiment protocol and timeline using gene chips, once we have defined and tested with RT-PCR of key genes. This will consist of two timepoints, with activated & unactivated samples (4 conditions). This will be run in triplicate, requiring 12 gene chips. In order to accomplish these objectives. Not only do we need supplies, we need personnel funding, which we do not yet have.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Millie Hughes-Fulford "Signal transduction and mechanical stress" Science’s STKE. 2004 Aug 31;2004 (249):RE12, , Aug-2004
Articles in Peer-reviewed Journals	Millie Hughes-Fulford, Eiko Sugano Thomas Schopper Chai-Fei Li J. B. Boonyaratanakornkit and Augusto Cogoli "Early immune response and regulation of IL-2 receptor subunits" In Press 2005 , Feb-2005
Articles in Peer-reviewed Journals	Hatton, J.P., Pooran, M., Li, C-F., Luzzio, C. Hughes-Fulford, M "A short pulse of mechanical force induces gene expression and growth in MC3T3-E1 osteoblasts via an ERK 1/2 pathway" J Bone Miner Res. 18(1), 58-66, Jan-2003 , Jan-2003
Articles in Peer-reviewed Journals	Millie Hughes-Fulford "Function of the cytoskeleton in gravisensing during spaceflight" Adv Space Res. 2003;32(8):1585-93. , Aug-2003
Books/Book Chapters	Millie Hughes-Fulford "Physiological effects of microgravity on osteoblast morphology and cell biology" in "Advances in Space Biology and Medicine" Ed. A . Cogoli COSPAR, Volume 8: 129-157 2003 , Sep-2003